

Selenium Full Notes

1. Introduction

Selenium is a Python automation tool used to control browsers for testing, scraping, and automation.

2. Installation

```
pip install selenium
```

3. Basic Setup

```
from selenium import webdriver

driver = webdriver.Chrome()
driver.get("https://example.com")
```

4. Finding Elements

```
driver.find_element("id", "username")
driver.find_element("class name", "form-control")
driver.find_element("xpath", "//input[@type='text']")
driver.find_element("css selector", "input.name")
```

5. Input and Click

```
driver.find_element("id", "username").send_keys("admin")
driver.find_element("id", "login").click()
```

6. Waits

```
driver.implicitly_wait(10)

from selenium.webdriver.common.by import By
from selenium.webdriver.support.ui import WebDriverWait
from selenium.webdriver.support import expected_conditions as EC

WebDriverWait(driver, 10).until(
    EC.presence_of_element_located((By.ID, "username"))
)
```

7. Navigation

```
driver.back()
driver.forward()
driver.refresh()
```

8. Extract Text and Attributes

```
text = driver.find_element("tag name", "h1").text
link = driver.find_element("tag name", "a").get_attribute("href")
```

9. Screenshot

```
driver.save_screenshot("page.png")
```

10. Headless Mode

```
from selenium.webdriver.chrome.options import Options

options = Options()
options.add_argument("--headless")

driver = webdriver.Chrome(options=options)
```

11. Advanced

```
driver.execute_script("window.scrollTo(0, document.body.scrollHeight);")

from selenium.webdriver.support.ui import Select
Select(driver.find_element("id", "country")).select_by_visible_text("Bangladesh")
```

12. Example Automation

```
from selenium import webdriver

driver = webdriver.Chrome()
driver.get("https://example.com/login")

driver.find_element("id", "username").send_keys("admin")
driver.find_element("id", "password").send_keys("1234")
driver.find_element("id", "login").click()
```

13. Best Practices

Use explicit waits, match driver version, and always close browser using `driver.quit()`.